Status	Finished
Started	Tuesday, 20 October 2026, 8:36 AM
Completed	Tuesday, 20 October 2026, 8:49 AM
Duration	12 mins 56 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The search stops
c. The list is sorted
d. The search continues in the left or right sublist ✓

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search
b. Merge search
c. Quick search
d. Linear search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Hashing
b. Dynamic search
c. Binary search
d. Linear search ✓

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Dividing the list in half repeatedly
b. Skipping every second element
c. Checking each element sequentially ✓
d. Sorting the list before searching

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search stops immediately ✓
c. The search continues until the end of the list
d. The search backtracks to find duplicate elements

Question **6**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. An error is raised
c. The last element is returned
d. The search is considered unsuccessful ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 89 and 99
b. 90 and 94
c. 89 and 94
d. 90 and 99 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list
b. Divide the list into two equal parts
c. Compare the target element with the middle element in the list ✓
d. Compare the target element with the first element in the list

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the left sublist ✓
c. At the beginning of the list
d. In the right sublist

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n/2
c. log n
d. n ✓

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(1)
c. O(n log n)
d. O👎 ✓

Question **12**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n ✓
c. n-1
d. n/2

Question **13**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search continues in the left sublist
b. The search continues in the right sublist
c. The list is sorted
d. The search ends successfully ✓

Question **14**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It can be applied only if the list is sorted ✓
b. It compares elements sequentially
c. It always starts from the beginning of the list
d. It works on unsorted lists

Question **15**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By dividing the list length by two ✓
c. By comparing each element sequentially
d. By starting from the first element